

AURAFLOW

Chinese Traditional Culture
Decision Assist Robot

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Abstract

This report documents the research and development of AURAFLOW, a decision-support robot grounded in the core logic of Xiao Liu Ren (a distinctive traditional Chinese divination method). Utilizing sensors to facilitate user interaction, the robot visualizes divination outcomes through the variable blooming amplitude of a mechanical flower and the modulation of light color and intensity. Additionally, results are displayed on a main interface to facilitate digital preservation for the user.

Team Division of Labor

Tanyang Li and Wenzhuo Yan are responsible for the software side, including coding and circuit planning. Zhiqi Yao, Jinge Yang, and Guoping Zhi handle the hardware design, including material selection and structural construction. The final assembly will be completed collaboratively by all team members. For key design milestones—such as core logic and visual direction—members will propose options and collectively decide on the final solution.



Core Logic

Xiao Liu Ren is a traditional Chinese divination method, attributed to Zhuge Liang and commonly described as “pinching the fingers to calculate.” Its mathematical structure makes it well-suited for computational translation.

In our system, three numbers are collected from user interactions, conceptually representing Tian Shi (Heaven’s Time), Di Li (Earth’s Advantage), and Ren He (Human Harmony). Using these numbers, the program performs a clockwise count beginning at the index-finger position on the left hand (Figure 1). Each subsequent count starts from the previous endpoint. The final landing point yields one of six outcomes—three auspicious and three inauspicious—each mapped to a corresponding Five Elements attribute, which then controls the flower’s motion and display lighting.

Software Component: Tanyang Li / Wenzhuo Yan

Input

We aimed to collect three parameters representing Heaven, Earth, and Human through sensors. Initial testing involved multiple modules, but unstable readings from the photoplethysmography sensor, weak GPS signal acquisition, and the standard Arduino’s inability to access real-time clock data led us to rule out several options. Concerns about limited interfaces also prompted preliminary dual-microcontroller communication tests, which proved ineffective.

With instructor guidance, the team switched to the Uno WiFi R4, enabling real-time clock and location retrieval via WiFi. For better interaction, an ultrasonic sensor was selected to convert hand distance into the Human parameter, improving user engagement.

Core Calculation

When the user presses the button, the system records the current time in seconds, the latitude and longitude, and the ultrasonic mapped distance (0-9). These represent the three numerical inputs for Heaven (Tian), Earth (Di), and Human (Ren). For multi-digit inputs (e.g., seconds, latitude, or longitude), all individual digits must be summed together to obtain the counting result (per Xiao Liu Ren rules). In the code, all digits are summed simultaneously, and then 2 is subtracted to obtain the final mapped number X (which accounts for the overlapping positions in the manual counting method). Since Xiao Liu Ren is a cycle of six results, $X \pmod{6}$ yields the final outcome. The code implementation uses string concatenation for the numbers, subtracts the ASCII value of '0' for numerical conversion, and then performs the calculation, resulting in the final required result (actual results range from -2 to 3, totaling six possible numbers), corresponding to the six Xiao Liu Ren outcomes.

```
int totalSum = 0;
// Concatenate Data
String allData = String(currentNum) + lat + lon + String(seconds);

// Sum all digits
for (int i = 0; i < allData.length(); i++) {
    if (isDigit(allData.charAt(i))) {
        totalSum += (allData.charAt(i) - '0'); // Char to Int
    }
}
// Formula: (Sum % 6) - 2
return (totalSum % 6) - 2;
```

Output

Upon the user's first button press, the Five Elements (Wu Xing) lights sequentially illuminate while data is recorded and processed, achieving a visual "thinking" process.

The user is then required to press the button a second time to view the final result. This two-step process is implemented based on the rationale that many people, at the moment they decide to seek divination, already hold an answer in their hearts and may not need to see the final outcome—much like the moment a coin is flipped but before it lands, the first thought that comes to mind is the true answer.

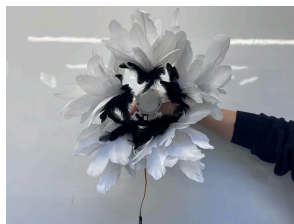
The second button press displays the result: based on the previously obtained number, the control state machine is activated; the corresponding attribute light illuminates; the mechanical flower opens to different angles driven by a torque motor (for auspicious results); and the stamen light adjusts its brightness to reflect the outcome. Simultaneously, the corresponding Xiao Liu Ren mnemonic verse is shown on the main screen in the form of a QR code.

Hardware Component: Zhiqi Yao, Guoping Zhi, Jing Yang

Based on earlier discussions, we adopted a flower-centered design driven by a servo, with five-element lights and a central stamen light as decoration. The hardware consists of two main parts: the flower structure and the "main plate," which holds the five-element lights.

Flower Structure

The flower frame is 3D-printed, with linkages connected to a servo to control opening and closing. Black-and-white feathers were chosen for petals to enhance soft light diffusion and echo Taiji color symbolism.



Main Plate

The plate contains a central opening for the flower, with the servo hidden below. Five-element lights are placed around it, and the base conceals wiring to evoke a root-like form. The graphic uses a layered Bagua-maze pattern.

The initial wooden design was replaced with acrylic due to fabrication difficulty and the need for CNC. Laser-cut acrylic required repeated size adjustments and small test cuts to manage heat sensitivity and avoid melting or dark edges. A stacked structure was used for depth, with a mirror acrylic sheet as the bottom layer to enhance visual impact.



Reflection

Code Logic

Insufficient early research made the code unnecessarily complex. Time was wasted on sensor and master-slave tests that WiFi on the main board could have handled. Deeper hardware understanding is needed.

Product Lifespan

The mismatch between the servo's circular motion and the linkage's linear motion reduced the 3D-printed base's durability. A gear-and-rack system should be used in future designs.

Assembly Process

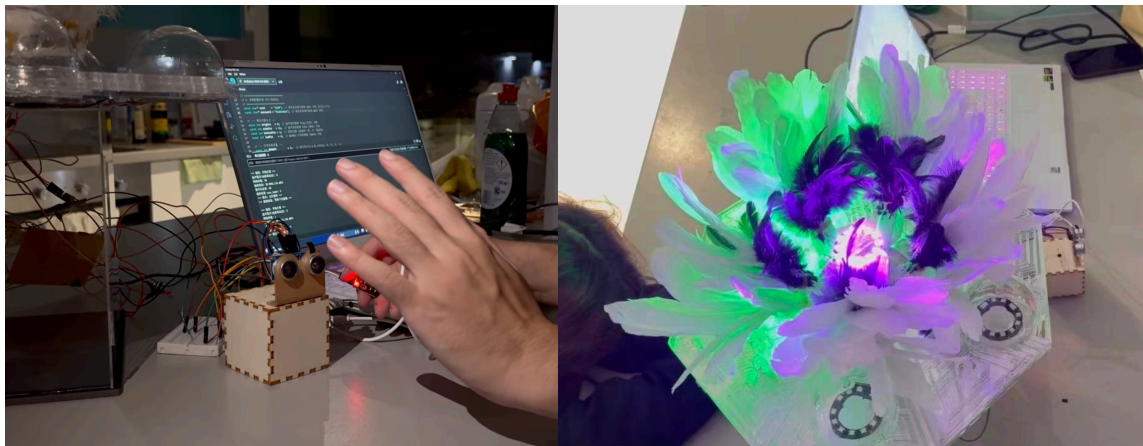
Lack of planning led to premature soldering and later wire cutting when parts didn't fit. Wire length issues and loose joints caused extra rework. A defined assembly sequence is needed.

Summary

For future improvements, our team will enhance coordination and align design methods earlier, ensuring clear steps from concept to logic and feasibility to anticipate risks sooner.

Technically, we will adopt a more structured testing workflow, including material trials, stress simulations, and planned routing to reduce rework.

Despite many challenges, this project emphasized the importance of teamwork, technical validation, and structural planning. These lessons will guide more efficient and reliable development in future projects.



Video Link: <https://youtu.be/r9fNzEMsPME>